

Lecture 2: Kirchhoff's Laws

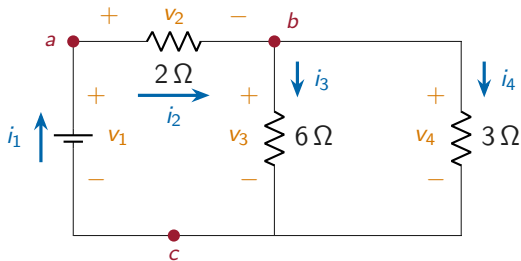
EE201 – Circuit Theory I

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■ current ■ voltage ■ power ■ topology

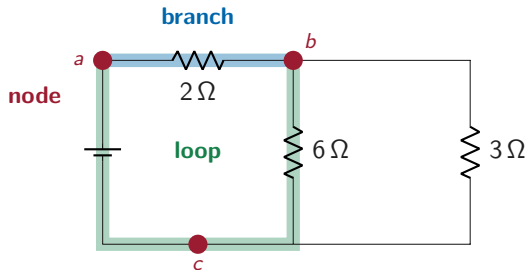
A circuit — and what we do not yet know



Four branches, each carrying a voltage and a current:

$$v_1, v_2, v_3, v_4 \quad \text{and} \quad i_1, i_2, i_3, i_4 \quad \implies \quad 2b = 8 \text{ unknowns}$$

Node, branch, loop



Node — a connection point; everything joined by ideal wire is *one* node, however long the drawing makes it. Here $n = 3$. **Branch** — one element, two nodes. Here $b = 4$. **Loop** — a closed path through no node twice.

What each element says — on its own



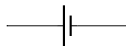
$$v = R i$$



$$v = L \frac{di}{dt}$$



$$i = C \frac{dv}{dt}$$



$$v = v_s(t)$$

For our circuit: $v_1 = 12 \text{ V}$, $v_2 = 2 i_2$, $v_3 = 6 i_3$, $v_4 = 3 i_4$

That is 4 equations for 8 unknowns. Exactly half —
and staring harder at the elements will never produce the rest.

Two kinds of equations — and they never mix

Terminal equations

$$v = Ri, \quad i = C dv/dt, \dots$$

b of them

what each element is
— *it does not know where it is*

+

Network equations

KCL and KVL

b of them

how they are wired
— *it does not know what they are*

Because KCL and KVL ignore the elements, they hold for **every** lumped circuit — linear or not, time-invariant or not, at DC or at 100 MHz.

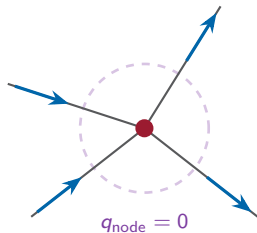
Kirchhoff's Current Law

$$\sum_{k \in \text{node}} \pm i_k(t) = 0$$

at every node, at every instant.

What flows in, flows out.

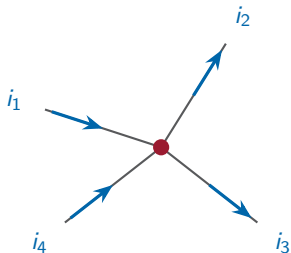
Why: a node cannot store charge



$$\frac{dq_{\text{node}}}{dt} = \sum_{\text{in}} i_k - \sum_{\text{out}} i_k = 0$$

Conservation of charge, plus *nothing piles up* — the first lumped assumption from Lecture 1.

Sign rule, and an example



Leaving positive:

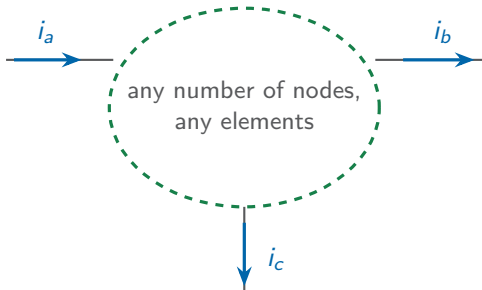
$$-i_1 + i_2 + i_3 - i_4 = 0$$

Given $i_1 = 3\text{ A}$, $i_2 = -2\text{ A}$, $i_3 = 6\text{ A}$:

$$-3 + (-2) + 6 - i_4 = 0 \implies i_4 = 1\text{ A}$$

A measured -2 A goes in *with its sign*. We never redraw the arrow.

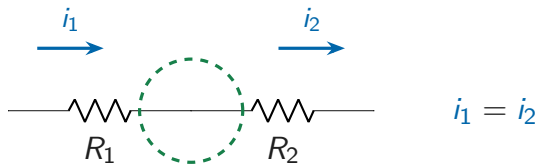
Nothing in the argument needed a point



$$-i_a + i_b + i_c = 0$$

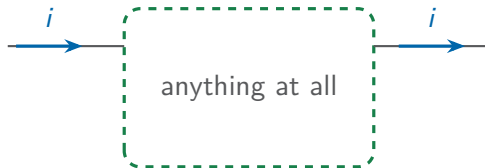
Only the branches that **cross** the surface appear. One node or a thousand, resistors or a whole amplifier — it makes no difference.

Consequence I: series means one current



Exactly two branches cross the surface. A chain of elements has **one** current — and that *is* the definition of “in series”.

Consequence II: a box gives back what it takes



This is what lets us say “**the** current through” a black box — and what makes a two-terminal element a meaningful idea at all.

Two things KCL is not

Not averaged

$\sum \pm i_k(t) = 0$ holds at each instant t *separately*. Nothing is averaged over time.

Not about v

It says nothing about voltage, resistance or power. Topology and current only.

Kirchhoff's Voltage Law

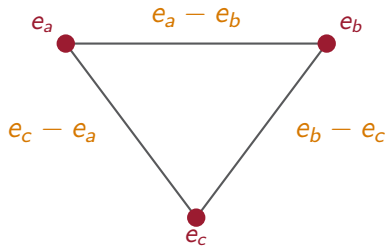
$$\sum_{k \in \text{loop}} \pm v_k(t) = 0$$

round **every loop**, at **every instant**.

Go all the way round and you come back to where you started.

As with KCL, this is the lumped model at work: it needs no changing magnetic field through the loop — assumption (b) from Lecture 1.

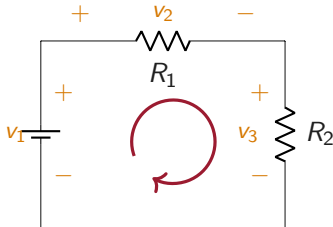
Why: the differences telescope



$$(e_a - e_b) + (e_b - e_c) + (e_c - e_a) = 0$$

Every node carries one number. A branch voltage is a *difference*. Around a loop they cancel in pairs.

Sign rule: walk the loop

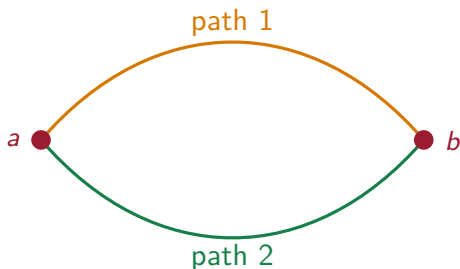


$$-v_1 + v_2 + v_3 = 0$$

- + when you traverse a branch from + to -;
- - when you go the other way.

Walking anticlockwise multiplies every term by -1 — same equation.

The same statement: voltage forgets the path

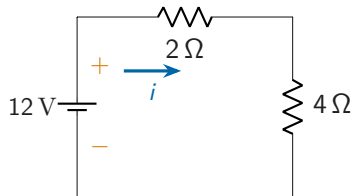


$$V_{ab} = V'_{ab}$$

Path 1 forwards then path 2 backwards is a loop, so the two must agree.

This is why “the voltage across R_2 ” means something — and why node potentials exist at all.

Example: one loop



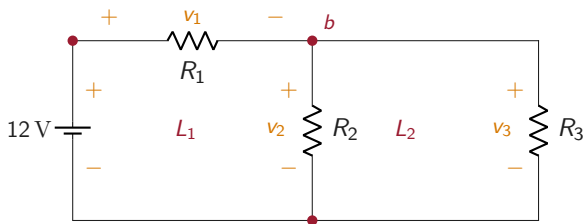
$$-12 + 2i + 4i = 0$$

$$\Rightarrow i = 2 \text{ A}$$

$$v_{2\Omega} = 4 \text{ V}, \quad v_{4\Omega} = 8 \text{ V}$$

$4 + 8 = 12$: the source's volts are shared in proportion to resistance.

Example: three loops, two equations

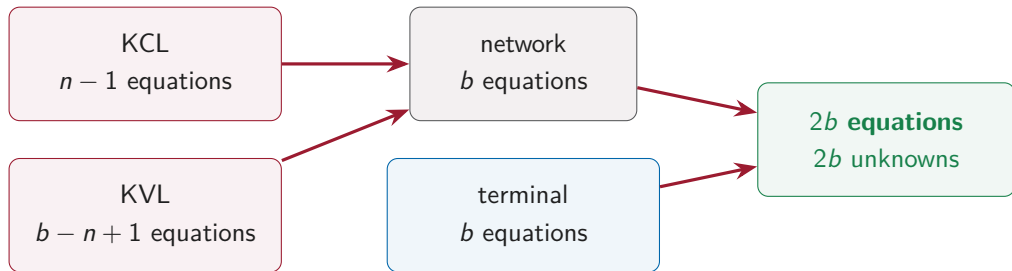


$$L_1 : -12 + v_1 + v_2 = 0 \quad L_2 : -v_2 + v_3 = 0 \quad \underbrace{L_3 : -12 + v_1 + v_3 = 0}_{= L_1 + L_2}$$

The third loop is **true but not new**. Only $\ell = b - n + 1 = 2$ of them count.

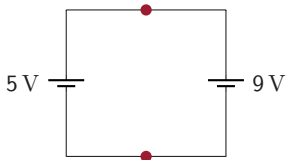
The whole bookkeeping, on one slide

n = number of **nodes**, b = number of **branches**. Here $n = 3$, $b = 4$.

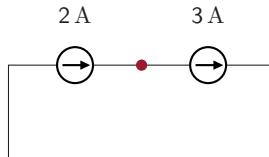


Why is one KCL redundant? Add up *all* the node equations: every branch appears once with $+$ and once with $-$, and the sum is $0 = 0$.

Some diagrams mean nothing at all



KVL: $5 = 9$

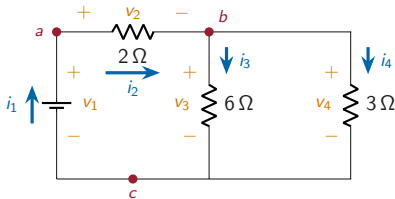


KCL: $2 = 3$

Both are **false whatever the other variables do** — the diagram describes no circuit. An ideal source promises to hold its value at any cost, and two such promises can disagree.

Give each source its internal resistance and the equations agree again, with a very large current. Sometimes that is harmless; sometimes it is why one of them makes smoke.

All 2b equations, once



terminal (4)

$$v_1 = 12 \quad v_3 = 6 i_3$$

$$v_2 = 2 i_2 \quad v_4 = 3 i_4$$

KCL (2)

$$a: -i_1 + i_2 = 0 \quad b: -i_2 + i_3 + i_4 = 0$$

KVL (2)

$$L_1: -v_1 + v_2 + v_3 = 0$$

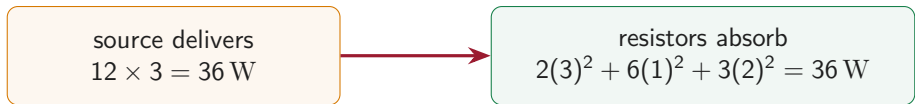
$$L_2: -v_3 + v_4 = 0$$

$$i_1 = i_2 = 3 \text{ A}, \quad i_3 = 1 \text{ A}, \quad i_4 = 2 \text{ A}$$

$$v_2 = v_3 = v_4 = 6 \text{ V}$$

Free check: power must balance

$$\sum_k p_k(t) = 0 \quad \text{in any circuit, at any instant}$$



Costs one line and catches most sign errors.

Summary

KCL	$\sum \pm i_k = 0$	at every node, every instant
KVL	$\sum \pm v_k = 0$	round every loop, every instant

- $2b$ unknowns = b terminal + $(n - 1)$ KCL + $(b - n + 1)$ KVL.
- KCL holds on any closed surface: elements joined end to end carry one current.
- KVL says voltage forgets the path: elements joined at both ends carry one voltage.
- Signs come from the reference arrows and polarities you drew — never from a guess about which way things really go.
- Neither law knows what the elements are — both survive nonlinear and time-varying circuits untouched.

Next lecture

What a two-terminal element can be

Independent voltage and current sources, the general definition of a resistor, and the five words for describing one — all read off a curve in the v – i plane.

Questions?

Notes for this lecture: see the course repository.